MICROSOFT GRAPH • MCP • INTUNE • ENTRA

Ask Intune and Entra in plain English

Microsoft Graph that doesn't hallucinate

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GraphMind • A grounded MCP server for Graph



Thank you, Sydney sponsors!

glueck kanja

DEVICIE

eG Innovations

The demo request sounds harmless

“Show me every noncompliant Windows device,
why it failed, and the last user who signed in.”

The risky part is not the English.
It is everything the model invents between intent and API execution.

01 Which resource family?

02 Which API version?

03 Which permissions?

04 Which properties and joins?

05 Read or write?

A fluent answer can still be a false route

STALE — Endpoint arrived after model training

PLAUSIBLE — A made-up path looks syntactically right

AMBIGUOUS — Intent spans multiple resources and versions



A 404 is obvious. A valid but wrong endpoint is worse.

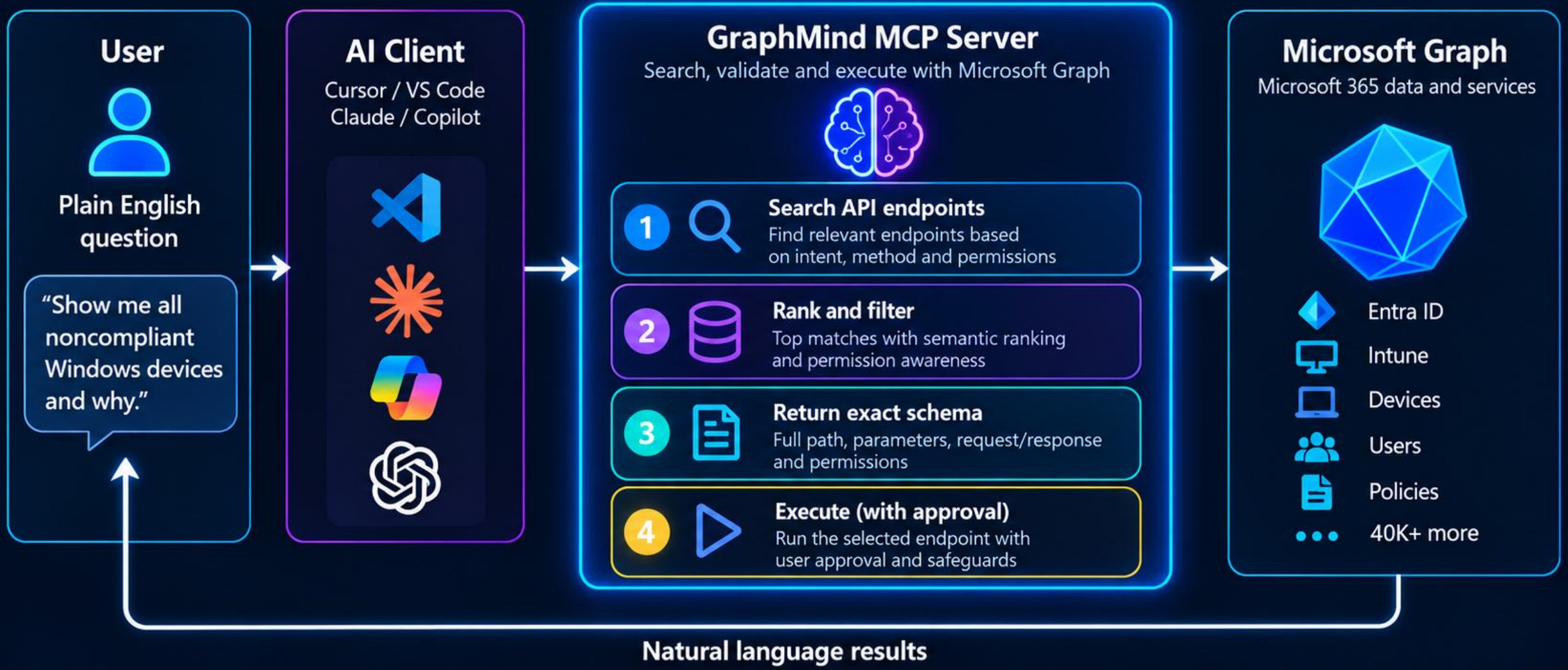
The project began with one uncomfortable question

1. **Observe** — Lokka made Graph easy to call
2. **Challenge** — But the model still supplied the path from memory
3. **Prototype** — Search the live spec before making the call
4. **Harden** — Add ranking, lifecycle awareness and write approval

The insight: MCP makes tools callable. Grounding makes them trustworthy.

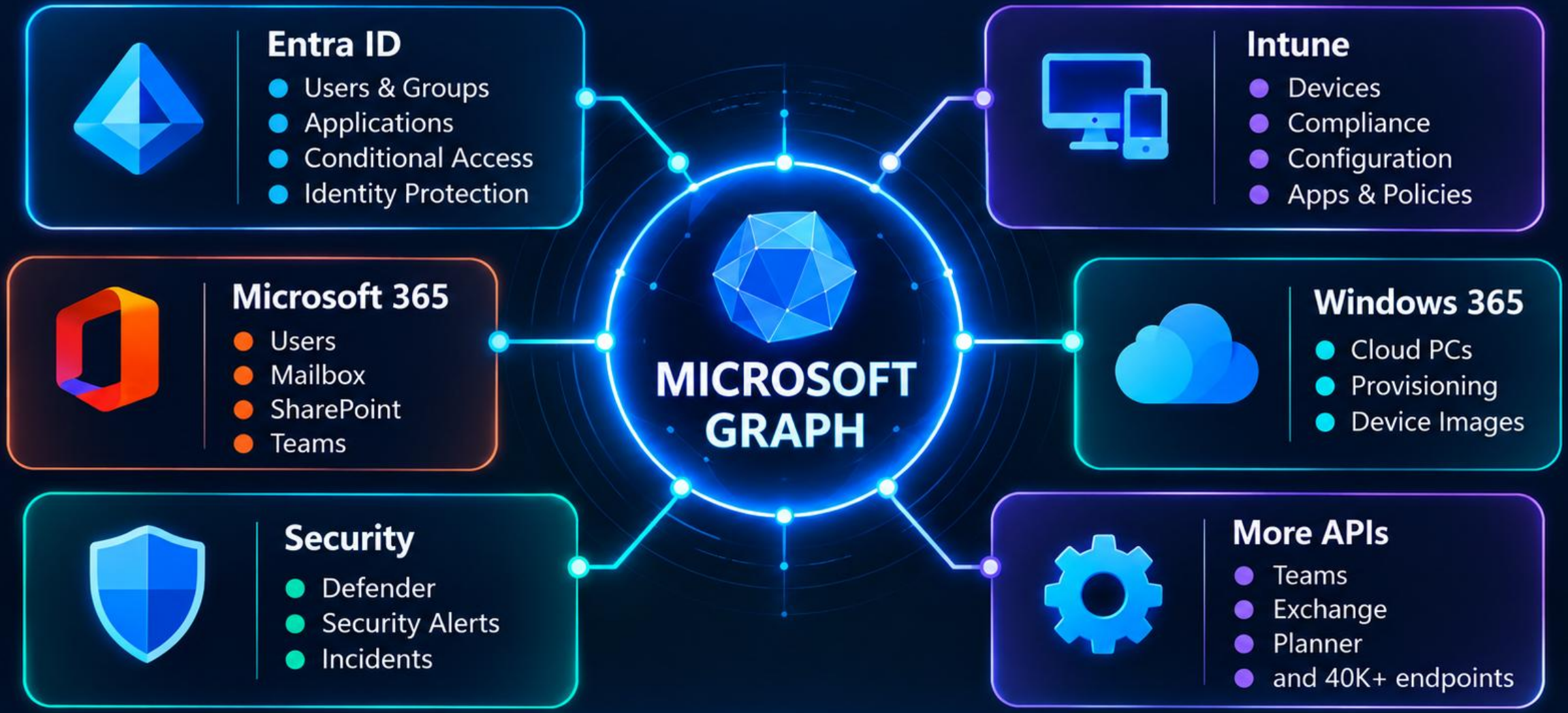
1 Architecture Overview

Natural language to Microsoft Graph with live schema lookup, validation and safe execution



Shows the end-to-end flow and positions GraphMind clearly in the ecosystem.

Supported Microsoft Services



Helps the audience see the breadth of what they can ask about.

2 Search and Ranking Flow

From natural language to the right API endpoint

A three-tier funnel that filters, reranks and enriches results to find the most relevant Microsoft Graph endpoint.



Visualises the three-tier funnel in more detail than the current slide.

3 Example Query to API

1 User question

"Show me all noncompliant Windows devices, why they failed, and the user who signed in."

2 GraphMind search results

1. /deviceManagement/managedDevices (v1.0)
2. /deviceManagement/deviceCompliance... (v1.0)
3. /auditLogs/signIns (v1.0)

3 Selected endpoint (schema)

GET /deviceManagement/managedDevices (v1.0)

Parameters | Response | Permissions

```
$filter =  'complianceState ne compliant'
$select =  'id,deviceName,complianceState,
            userPrincipalName,lastSyncDateTime'
$expand =  'user,deviceCompliancePolicy'
```

4 Result (natural language)

Found 12 noncompliant Windows devices.
Most failed due to missing encryption policy.
Last sign-in details included.

A concrete before-and-after example builds credibility and makes it real.

5 Safety and Guardrails

Prevent risky actions and enforce governance before any operation runs.

Before execution

- Endpoint exists in current spec
- Required permissions validated
- Read vs write operation detected
- User approval for write actions
- Scopes and tenant boundaries

Write operation?

Create, update or delete?

Review details

Confirm to proceed

Execution controls

- Read-only by default
- Explicit confirmation for write operations
- No execution without valid schema
- Audit and logging
- Permission scoping

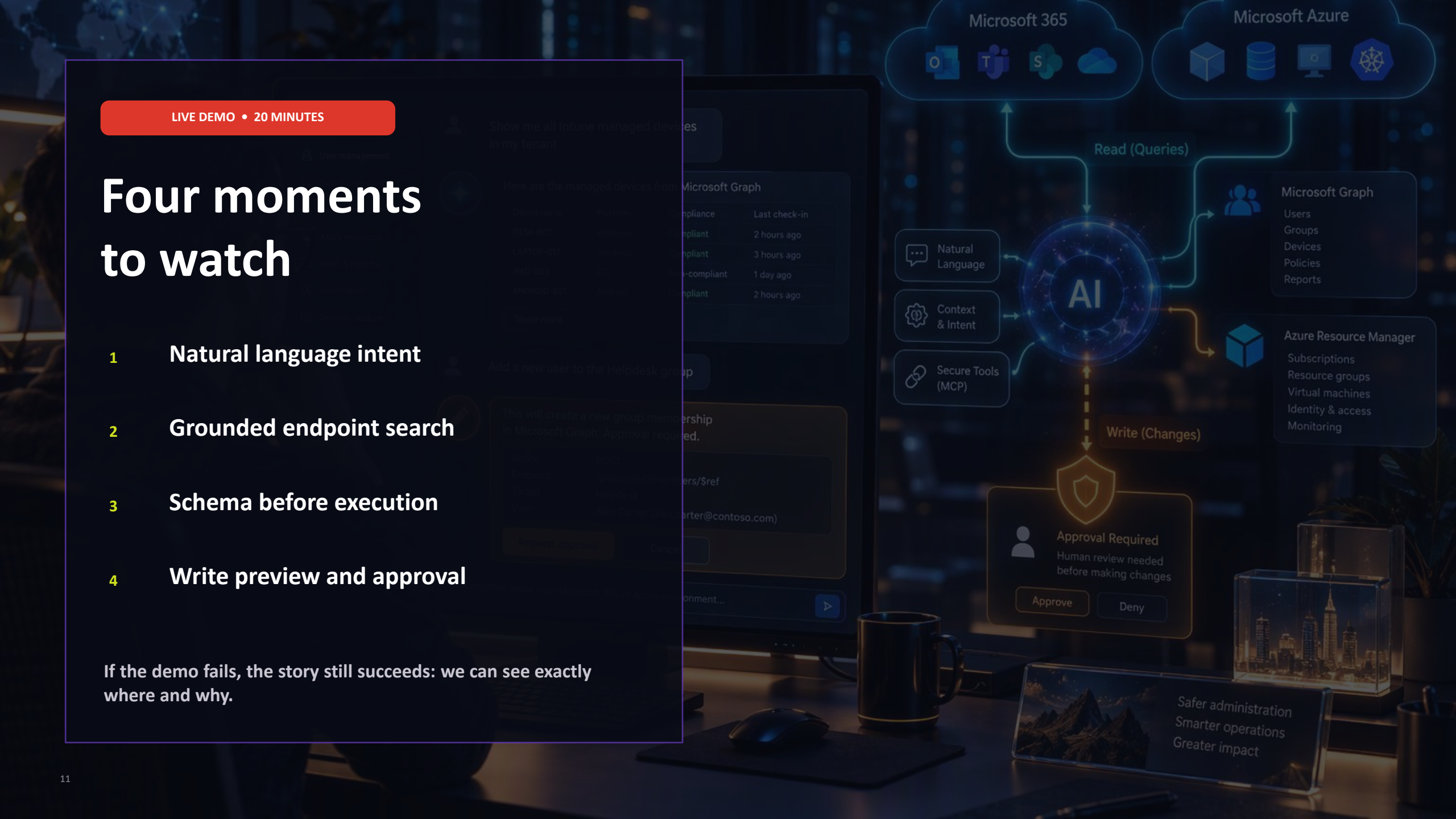
Builds trust by showing how *GraphMind* reduces risk.

LIVE DEMO • 20 MINUTES

Four moments to watch

1. Natural language intent
2. Grounded endpoint search
3. Schema before execution
4. Write preview and approval

If the demo fails, the story still succeeds: we can see exactly where and why.



Demo runbook: one read, one investigation, one guarded write

01	INTUNE INVENTORY	List noncompliant Windows devices and return a small sample.	Read + pagination
02	ENTRA INVESTIGATION	Find Conditional Access policies and show the selected endpoint schema.	Search + explain
03	SAFE CHANGE	Prepare a group membership change, but stop at the approval preview.	Write guardrail

Backup plan: use captured output and continue the endpoint-selection story.

What GraphMind does, and what it deliberately does not

DOES

- ✓ Ground endpoint choice in the current spec
- ✓ Rank likely matches for natural-language intent
- ✓ Expose version, schema and write confirmation
- ✓ Keep credentials and execution local

DOES NOT

- × Guarantee the user's intent is correct
- × Replace least-privilege permissions
- × Make beta APIs production-safe
- × Remove the need for testing and audit

The takeaway

Let the model interpret intent.
Make the system verify the path.

01 Search before call

02 Prefer evidence over memory

03 Preview before write

github.com/askaresh/graphmind

Thank you • Questions

SHARING KNOWLEDGE. TOGETHER.